Resolved referee R042 [Gap; revised] – The compactified closed-graph criterion.
Original source line 1117. The parenthesis appears intended to impose the correct compactified-target hypothesis; make it explicit by taking B to be a closed definable subset of $(\mathbb{P}^1(\mathrm{VF}))^m$, and take graph closure in $A \times B$. The example $f(0) = 0$, $f(x) = 1/x$ for $x \neq 0$ has closed graph in affine VF^2 but is discontinuous at 0: this is a counterexample only to the unbounded affine-target version, not to the intended compactified statement. For the reverse implication, apply curve selection to the set of input points x with $f(x) \notin \mathfrak{b}$, approaching a; apply the finite-limit lemma to the resulting finite-valued image in B. A limit point b' lies outside the clopen neighborhood $\mathfrak{b}$, and (a, b') lies in the graph closure. Also, "projective line with the valuation topology" is more precise than a literal topological one-point compactification; local compactness is not assumed here.

Author resolution. The author approves making the intended projective ambient space explicit: the affine target is B, and graph closure is in $A \times B^{\mathrm{p}}$, with the valuation topology. The reverse proof selects bad inputs first and then takes a finite projective limit of their images. Temporary base enlargement supplies curve selection without changing the geometric conclusion or imposing global VF-generation.

Independent review of the original proofs. The two proofs below remain exactly as in the uploaded main text. The report's arity request in R043 is withdrawn: the explicit subsets of VF are already unary. The singleton limit is supported by the cited prequel's finite-limit cardinality bound. The whole-annulus equality uses eventual radial valuation laws for the map and its inverse; the inverse supplies the reverse inclusion. This is an implicit justification, not a counterexample to the statement. For R044, the partition property gives the lower bound directly; its limit-point contradiction is valid. The last uniformity step uses definable fiber bounds, as in the author's earlier induction. The synchronized finite choices need the model/compactness justification over a temporary base already used in the preceding approved geometric proofs. These observations do not establish that the original proofs require wholesale replacement. The earlier long green rewrites remain unapplied and are held for reassessment.

Lemma 2.51. *Let $f : A \longrightarrow B$ be a definable bijection between two open subsets of* VF*. Suppose that B is bounded. Then f is weakly concentric, that is, for every $a \in A$ there is a $b \in \mathrm{VF}$ such that, for every sufficiently large $\gamma \in \Gamma$, $f(\mathfrak{o}(a, \gamma) \smallsetminus a) = \mathfrak{o}(b, \delta) \smallsetminus b$ for some $\delta \in \Gamma$.*

Note that, compared to Lemma 2.32, the assumption of this lemma is stronger and its conclusion weaker, but the point is that we do not need a partition to achieve it.

Proof. By Lemma 2.45, for any $a \in A$, $\lim_{A \to a} f$ exists in the closure of B, which we denote by b_a. By C-minimality and o-minimality in the Γ-sort, $\mathfrak{c}(b_a, \delta) \smallsetminus \mathfrak{o}(b_a, \delta) \subseteq B$ for all sufficiently large $\delta \in \Gamma$. Also, for all sufficiently large $\gamma \in \Gamma$ there is a $\delta \in \Gamma$ such that

$$f(\mathfrak{c}(a, \gamma) \smallsetminus \mathfrak{o}(a, \gamma)) = \mathfrak{c}(b_a, \delta) \smallsetminus \mathfrak{o}(b_a, \delta)$$

The lemma follows. □

Partly resolved referee R043 [Gap; revised] – The annulus equality needs a proof.
Original source line 1132. Specify that A and B each have one VF-coordinate. The finite limit is a singleton $\{b\}$ by [17, Lemma 9.4]; nonemptiness follows because the values are nonempty on a punctured neighborhood. The equality of whole thin annuli needs more explanation. Here is a repair. The valuation calculation in the proof of [17, Lemma 8.3], applied to $x \mapsto \mathrm{val}(f(x) - b)$, gives

$$\mathrm{val}(f(x) - b) = q\,\mathrm{val}(x - a) + c \qquad (q \in \mathbb{Q}_{>0})$$

on a sufficiently small punctured neighborhood: each unary polynomial has eventually monomial valuation, and the limit forces positive slope. Remove the at most one exceptional input with $f(x) = b$. By C-minimality, B contains a punctured neighborhood of b. The inverse has a unique projective limit there, necessarily a, since points $x \longrightarrow a$ give $f(x) \longrightarrow b$. Apply the same valuation calculation to the inverse. The resulting two affine maps of tail radii are inverse, so every sufficiently small target shell consists exactly of

images of its corresponding source shell. Taking their unions proves the asserted punctured-disc equality. Thus the lemma is repairable; an additional assertion that $q = 1$ is unnecessary.

Author resolution. The arity objection is withdrawn: the statement already says that A and B are subsets of VF, so both are unary. The general tuple convention does not alter that explicit scope, as the author confirms. No arity amendment or change to the statement is needed. The separate proof-detail comments are being assessed against the paper's own argument; no proposed proof replacement has been approved or applied.

Definition 2.52. Let $p : A \longrightarrow \Gamma$ be a definable function. We say that p is an *$\mathfrak{o}$-partition* of A if, for every $a \in A$, the function p is constant on $\mathfrak{o}(a, p(a)) \cap A$.

This makes sense even when A has no VF-coordinates, in which case any definable function $A \longrightarrow \Gamma$ is an $\mathfrak{o}$-partition of A.

Lemma 2.53. *Let p be an $\mathfrak{o}$-partition of A. Suppose that A_{RV} is doubly bounded and A_t is closed and bounded for every $t \in A_{\mathrm{RV}}$. Then $p(A)$ is doubly bounded.*

Proof. We first handle the case that A has no RV-coordinates. Since A is closed and bounded, $p(A)$ must be bounded from below. Suppose for contradiction that $p(A)$ is not bounded from above. For each $\gamma \in \Gamma$, let $A_\gamma = \{a \in A \,|\, p(a) > \gamma\}$. Choose a definable continuous finitary function, that is, a curve

$$f : \mathcal{M} \smallsetminus 0 \longrightarrow \mathcal{P}\Big(\bigcup_{c \in \mathcal{M} \smallsetminus 0} c \times A_{\mathrm{val}(c)} \Big)$$

such that $f(c)_1 = c$. Write $A_c = f(c)_{>1}$. So $p(A_c) > \mathrm{val}(c)$ for all $c \in \mathcal{M} \smallsetminus 0$. Since A is closed, the limit points of f are all of the form $(0, e)$ with $e \in A$. So there is a $c \in \mathcal{M} \smallsetminus 0$ with $\mathrm{val}(c) > p(e)$ such that $A_c \cap \mathfrak{o}(e, p(e)) \neq \emptyset$. Since $p(a) > \mathrm{val}(c)$ for all $a \in A_c$, this contradicts the assumption that p is an $\mathfrak{o}$-partition of A. So $p(A)$ is bounded from above.

It is easy to see that the general case follows from this special case and Lemma 2.41. □

Referee R044 [Gap] – Bounds for an open-disc partition.
Original source line 1152. The lower bound has a short direct proof: choose η with $A \subseteq \mathcal{O}_\eta^n$. If $p(a) < \eta$ for some a, then $A \subseteq \mathfrak{o}(a, p(a))$, forcing p to be constant; otherwise $p(A) \subseteq [\eta, \infty)$. For the upper-bound argument, justify the nonempty finite choices in $A_{\mathrm{val}(c)}$ using the corrected curve-selection/model argument, and use the nonempty finite limit supplied by the limit lemma. In the general case, take a definable bound for each fiber image $p(A_t)$ by its definable endpoints in Γ, then apply Lemma 2.41; fiberwise boundedness by itself is not the uniform conclusion.

This lemma, in its various incarnations, is crucial for the good behavior of motivic Fourier transform (see [8, § 11] and [20]). For essentially the same reason, the main construction of this paper depends heavily on it (often in combination with Lemma 2.41).

Lemma 2.54. *Let $A \subseteq \mathrm{VF}^n$ be a definable set. Suppose that, for each $a \in A_1$, the fiber $A_a \subseteq \mathrm{VF}^{n-1}$ is open (respectively, closed). Then there is a finite definable set $A' \subseteq A_1$ such that $A \cap ((A_1 \smallsetminus A') \times \mathrm{VF}^{n-1})$ is open (respectively, closed) in $(A_1 \smallsetminus A') \times \mathrm{VF}^{n-1}$.*

Proof. Since either one of the two cases implies the other, we shall just concentrate on the open case. Consider the definable set

$$S = A \cap \mathrm{Cl}((A_1 \times \mathrm{VF}^{n-1}) \smallsetminus A),$$

where Cl is the closure operator in $A_1 \times \mathrm{VF}^{n-1}$; in other words, S is the boundary of A minus the frontier of A. It is enough to show that S_1 is finite. To that end, suppose for contradiction that S_1 contains a disc.